Economics of Organizations and Incentives

MPE – 4th term, 2026

Tutoring: paper presentations

Maheshri and Mastrobuoni (2021)

Görges (2021)



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Today

Part I
Maheshri, V. and Mastrobuoni, G. (2021). The Race Between Deterrence and Displacement: Theory and Evidence from Bank Robberies. *The Review of Economics and Statistics*, 103(3), 547-562.

Part II
Görges, L. (2021). Of housewives and feminists: Gender norms and intra-household division of labour. *Labour Economics*, 72, 102044.

For each paper: motivation, model, a case in pairs, data, results and critique.

Part I. The Race Between Deterrence and Displacement

Motivation

- Households and firms spend over a trillion dollars annually on security measures to prevent crime. These private expenditures complement similarly sized public expenditures.
- A large literature estimates the effectiveness of these measures in deterring crime, and its findings are used to allocate law enforcement resources and to evaluate private security.
- An important side effect of any criminal deterrent is its potential to **displace** crime: the effect of a security measure in one unit on crime in neighboring units, where neighbors may be defined along time, space or type of crime.
- Displacement does not affect the private benefits of deterrence, but it may lead to an overestimate of its **social** benefits, prompting a public intervention.

Motivation: the Brazilian context

- Public security expenditure reached R$ 153 billion in 2024.
- Private security corresponded to 0.94% of GDP, against 1.35% for public security (2015).
- There are about 520 thousand formal private security guards (2024), against about 480 thousand military and civil police officers (2020).
- Banks spend about R$ 9 billion a year on security, and federal law requires at least two guards in every bank branch (Lei 14.967/2024, art. 33).

Sources: Anuário Brasileiro de Segurança Pública (2024, 2025), SAE/Presidência da República (2018), MJSP (2022), Febraban (2022), Lei 14.967/2024.

Identifying displacement: empirical challenges

1. **Reverse causality.** Crime is determined in an equilibrium between potential criminals and potential targets, so investments in crime prevention reflect the underlying propensity for crime.
2. **Strategic interaction among targets.** The vulnerability of one target is a function of the vulnerability of alternative targets. Banks invest in response to the decisions of other banks, which creates multicollinearity and simultaneity problems.
3. **Contamination.** Units are not well defined a priori. A patrol on one block may also deter crime on the neighboring block (Barr and Pease, 1990).
4. **Measurement error** in crime data.

Related literature and contribution

- **Criminology.** Braga (2005) and Bowers et al. (2011) find little displacement and some diffusion of benefits, but areas of displacement are likely misspecified.
  → *This paper observes all potential targets and their exact locations, and varies the market definition instead of assuming it.*
- **Police and crime.** Di Tella and Schargrodsky (2004), Klick and Tabarrok (2005) and Draca, Machin and Witt (2011) lack the power to measure displacement (Donohue, Ho and Leahy, 2013).
  → *Precise estimates of displacement from a decade of data on all bank robberies.*
- **Private precaution.** Unobservable devices reduce thefts for all (Ayres and Levitt, 1998). Observable devices divert thefts to unprotected targets (Gonzalez-Navarro, 2013, van Ours and Vollaard, 2016).
  → *A model showing that displacement can create a coordination game among targets, so the direction of policy is ambiguous.*

This paper

Question: do security guards displace bank robberies to neighboring banks, and what does this imply for policy?

- **Data:** all Italian bank branches, 2000 to 2009 (annual risk of attempted robbery of 7%).
- **Theory:** displacement may generate a coordination game among banks.

Main results

- A guard reduces the probability of a robbery by 2.7 to 4.4 pp (31% to 50%).
- Half of the deterred robberies are displaced to nearby unguarded banks, and none to guarded banks.
- Restricting guards in rural markets and requiring them in cities could be socially beneficial.

A simple model of displacement

Setup

- Banks $i = 1, \ldots, N$ operate in a single market, defined as the set of banks viewed as substitutes by potential robbers.
- Each bank chooses whether to hire a guard, $g_i \in \{0, 1\}$, at cost $c_i > 0$. The expected loss in the event of robbery is $L_i > 0$.
- Probability of being robbed: $p(g_i, g_{-i})$, where g_{-i} is the number of neighboring banks that hire guards.

$$p(0, g_{-i}) - p(1, g_{-i}) \geq 0 \qquad \text{deterrence} \qquad (1)$$

$$\frac{\partial p}{\partial g_{-i}} \geq 0 \qquad \text{displacement} \qquad (2)$$

$$\left.\frac{\partial p}{\partial g_{-i}}\right|_{g_i=0} - \left.\frac{\partial p}{\partial g_{-i}}\right|_{g_i=1} \geq 0 \qquad \text{differential displacement} \qquad (3)$$

The hiring decision

Bank i hires a guard if its expected loss with a guard, including the hiring cost, is lower than without one:

$$p(1, g_{-i})L_i + c_i < p(0, g_{-i})L_i \tag{4}$$

or, equivalently,

$$\underbrace{p(0, g_{-i}) - p(1, g_{-i})}_{\pi(g_{-i})} > \underbrace{\frac{c_i}{L_i}}_{\lambda_i} \tag{5}$$

- $\pi(g_{-i})$, the **guard premium**: marginal benefit of a guard in units of expected robberies. It varies only with market conditions.
- λ_i: marginal cost of a guard in units of expected robberies. It varies with i, so banks can be ordered as $0 < \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_N$.
- Under (3), the guard premium is nondecreasing in g_{-i}.

Equilibrium

An equilibrium is a set of hiring decisions by all banks such that no bank would benefit from unilaterally deviating. Since bank i hires only if all banks $j < i$ do, the equilibrium is determined by the marginal bank.

Proposition 1

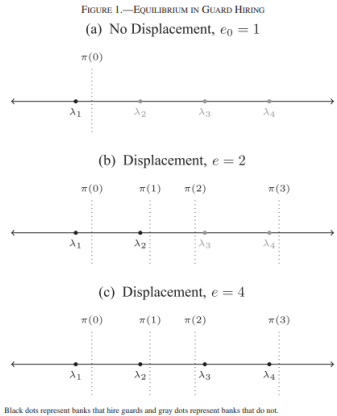
Suppose p satisfies (1) to (3) and exhibits a positive level of deterrence. Define $\lambda_{N+1} = \infty$. Then

(a) *(No displacement)* If (2) holds with equality, $\pi(g_{-i}) \equiv \pi(0)$ and e_0 banks hire guards in equilibrium, where e_0 uniquely satisfies $\lambda_{e_0} < \pi(0)$ and $\lambda_{e_0+1} \geq \pi(0)$.

(b) *(Existence)* In equilibrium, $e \geq e_0$ banks hire guards for any e such that $\lambda_e < \pi(e-1)$ and $\lambda_{e+1} \geq \pi(e)$.

(c) *(Uniqueness)* Let i be the smallest positive number such that $\pi(i-1) \leq \lambda_i$. For all $j > i$ such that $\lambda_j < \pi(j-1)$, either $i-1$ or j banks may hire guards in equilibrium.

Equilibrium in guard hiring (Figure 1)



- **Panel a.** Deterrence without displacement: the guard premium is constant at $\pi(0)$, and banks with $\lambda_i < \pi(0)$ hire guards.
- **Panel b.** With displacement, the premium increases with the number of guards, and two banks hire guards.
- **Panel c.** Since λ_3 and λ_4 lie between $\pi(2)$ and $\pi(3)$, a second equilibrium with four guards exists. Banks 3 and 4 play a coordination game.

Source: Maheshri and Mastrobuoni (2021), Figure 1

Welfare

- Displacement is a negative externality ignored by banks, so basic intuition suggests that an unregulated market features too many guards.
- This intuition is flawed, since displacement may create coordination games among banks.

Proposition 2

The socially optimal number of guards in a market may be higher or lower than the number of guards hired in a competitive equilibrium.

- In Figure 1, if the socially optimal number of guards is three, one equilibrium features too many guards and the other too few.
- It is not obvious whether regulation should encourage or discourage the hiring of guards, even though guards generate negative externalities.

Complete coordination and policy

Proposition 3 (complete coordination)

(a) If $\pi(0) < \lambda_1$, an equilibrium exists in which no bank hires a guard.

(b) If $\pi(N) > \lambda_N$, an equilibrium exists in which all banks hire guards.

- Complete coordination arises when banks are similar, deterrence is relatively low and displacement is relatively high.
- **Price regulations** (Pigouvian taxes or subsidies) operate through c_i and shift all λ_i. They are ill suited to fix coordination problems.
- **Quantity regulations** (guard requirements or restrictions) eliminate the coordination problem by forcing all banks to one equilibrium.

In-class exercise

Exercise: a market with two banks

Two neighboring branches, A and B, simultaneously decide whether to hire a guard (G) or not (N). The annual probability that a branch is robbed, $p(g_i, g_{-i})$, is

		Neighbor N	Neighbor G
Own branch	N	0.10	0.12
	G	0.06	0.06

Each robbery costs the bank $L =$ R$ 500 thousand and a guard costs $c =$ R$ 25 thousand per year. Each bank minimizes its expected loss $pL + cg$.

Exercise: questions

(a) Represent the game in normal form, with payoffs equal to minus the expected loss (in R$ thousand).

(b) Compute the guard premia $\pi(0)$ and $\pi(1)$ and the marginal cost λ. Verify that p satisfies assumptions (1) to (3).

(c) Find all pure-strategy Nash equilibria and the mixed-strategy equilibrium. Relate your answer to Proposition 3.

(d) Suppose there is no displacement: a neighbor's guard does not change a branch's probability of being robbed, so that $p(N, G) = p(N, N) = 0.10$ and $p(G, G) = p(G, N) = 0.06$. Characterize the equilibrium using Proposition 1(a).

(e) Each robbery imposes an additional cost H on society. For which values of H does the equilibrium with two guards have a lower social cost than the equilibrium with none? Relate your answer to Proposition 2.

Data

Data

- Yearly **Census of Bank Branches** of the Italian Banking Association (ABI), 2000 to 2009, with the precise location (latitude and longitude) of each branch.
- Full history of attempted robberies, reported to OSSIF, the ABI research center on crime against banks. Coverage: 71% of all robberies, and 83% between 2004 and 2009.
- Full history of investments in 37 security measures, including guards, bulletproof glass, security vestibules and time locks.
- Markets are defined by dividing Italy into squares of equal area, from 800 km down to 250 m.

Summary statistics (Table 1)

	All	Provinces, robberies	
		Below median	Above median
Number of robberies	0.07 (0.30)	0.04 (0.22)	0.10 (0.36)
Guard	0.08 (0.27)	0.04 (0.19)	0.13 (0.33)
Number of security devices	7.48 (4.94)	7.59 (5.22)	7.36 (4.61)
No substitute in 500 m market	0.41 (0.49)	0.46 (0.50)	0.36 (0.48)
Branch-years	245,712	125,401	120,311

Means, with standard deviations in parentheses.

The spatial distribution of guards mimics that of robberies, which anticipates the endogeneity problems discussed above.

Descriptive evidence of coordinated hiring (Figure 5)

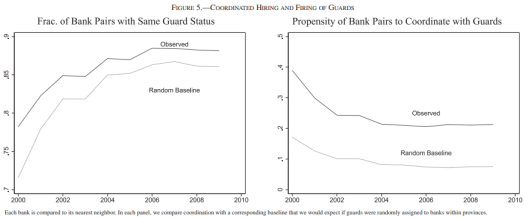


FIGURE 5.—COORDINATED HIRING AND FIRING OF GUARDS

Each bank is compared to its nearest neighbor. In each panel, we compare coordination with a corresponding baseline that we would expect if guards were randomly assigned to banks within provinces.

- Each bank is compared with its nearest neighbor, against a random assignment baseline.
- In 2000, over 40% of pairs with at least one guard had both banks guarded. With random hiring, the share would be about 20%.
- The gap of about 20 pp is persistent over the decade.
- Suggestive, though not dispositive, of coordination.

Source: Maheshri and Mastrobuoni (2021), Figure 5

Empirical strategy

Estimating equation

For bank i in market j and year t:

$$r_{ijt} = \beta_1 g_{ijt} + \beta_2 g_{-ijt} + \beta_3 g_{ijt} g_{-ijt} + \epsilon_{ijt} \tag{6}$$

- r_{ijt}: robbery attempt. g_{ijt}: bank i has a guard. g_{-ijt}: fraction of the other banks in market j with guards, so that displacement does not depend on market size.
- β_1: deterrence. β_2: displacement to unguarded banks. β_3: differential displacement between guarded and unguarded banks. In the model, $\pi(0) = -\beta_1$ and $\pi(N) = -(\beta_1 + \beta_3)$.
- **Endogeneity:** unobserved determinants of robbery in ϵ_{ijt} are correlated with the hiring decision of bank i, since p enters the hiring condition (4), and with the decisions of its neighbors. In the language of Manski (1993), β_2 and β_3 correspond to correlated effects.

Identification: successively smaller markets

- Estimate the simple deterrence regression with market and year fixed effects, for squares from 800 km down to 250 m:

$$r_{ijt} = \beta_1 g_{ijt} + \lambda_j + \lambda_t + \epsilon_{ijt} \tag{7}$$

- The deterrent effect of a guard should not depend on the size of the market. If β_1 is stable below some market size, the fixed effects control for environmental confounders.
- Limiting case: bank fixed effects. Guards are observed only at the end of each year, so years in which a guard is hired or fired are excluded.
- The full model (8) adds $\beta_2 g_{-ijt}$ and $\beta_3 g_{ijt} g_{-ijt}$ to (7). Displacement estimates could then be biased only by confounders that vary over time within markets and across markets within a year, and that are uncorrelated with the confounders of deterrence.

Results

Deterrence in successively smaller markets (Table 2)

	Fixed effects	Deterrent effect	# FE
(1)	None	-0.0043 (0.0041)	0
(2)	800 km	-0.0068^{*} (0.0040)	6
(3)	400 km	-0.0142^{***} (0.0037)	12
(4)	200 km	-0.0193^{***} (0.0036)	28
(5)	100 km	-0.0275^{***} (0.0035)	74
(6)	50 km	-0.0345^{***} (0.0036)	211
(7)	25 km	-0.0402^{***} (0.0037)	651
(8)	10 km	-0.0426^{***} (0.0038)	2,773
(9)	5 km	-0.0441^{***} (0.0038)	5,644
(10)	2 km	-0.0430^{***} (0.0041)	9,509
(11)	1 km	-0.0421^{***} (0.0040)	12,748
(12)	500 m	-0.0386^{***} (0.0042)	16,775
(13)	250 m	-0.0338^{***} (0.0044)	21,059
(14)	Bank	-0.0157^{***} (0.0040)	33,672
(15)	Bank, no switches	-0.0366^{***} (0.0064)	31,077

Dependent variable: number of robberies. Year fixed effects in all regressions. Standard errors clustered by 50 km squares.

- Without fixed effects, deterrence and endogenous investment cancel out.
- Estimates stabilize at about 4 pp between 25 km and 500 m.
- Bank fixed effects amplify the misclassification of hiring dates. Excluding switching years restores the estimate.

Deterrence and displacement (Table 3)

Market size	50 km	25 km	10 km	5 km	2 km	1 km	500 m	250 m
Guard (β_1)	-0.0348^{***}	-0.0373^{***}	-0.0391^{***}	-0.0407^{***}	-0.0382^{***}	-0.0369^{***}	-0.0318^{***}	-0.0289^{***}
	(0.0053)	(0.0054)	(0.0051)	(0.0048)	(0.0048)	(0.0044)	(0.0045)	(0.0051)
% neighbors with guards (β_2)	0.0020	0.0063	0.0058	-0.0001	0.0109	0.0181^{**}	0.0169^{**}	0.0147^{*}
	(0.0271)	(0.0236)	(0.0149)	(0.0118)	(0.0091)	(0.0082)	(0.0078)	(0.0081)
Guard $\times$ % neighbors (β_3)	0.0014	-0.0148	-0.0171	-0.0171	-0.0231	-0.0200	-0.0336^{***}	-0.0260^{**}
	(0.0286)	(0.0271)	(0.0231)	(0.0192)	(0.0145)	(0.0130)	(0.0119)	(0.0131)
p-value ($\beta_2 + \beta_3 = 0$)	0.920	0.781	0.615	0.351	0.339	0.863	0.113	0.348
Average number of neighbors	357	170	65.6	32.8	11.0	5.0	2.5	1.5

Dependent variable: number of robberies. Square and year fixed effects. Standard errors clustered by 50 km squares in parentheses.

- Deterrence estimates are nearly identical to those of Table 2.
- Displacement varies with the market definition: it appears only in markets of 1 km or less, where banks are closer substitutes.

Interpretation and robustness

- In markets smaller than 1 km^2, displacement to unguarded banks is 1.5 to 2 pp: if an unguarded bank's neighbors hire guards, its probability of being robbed increases by roughly 20%.
- No statistically significant displacement to guarded banks ($\beta_2 + \beta_3$), even in the smallest markets.
- Robust to market-specific trends, other security devices, the pre-2008 sample and lagged robberies (Tables 4 and 5).

Summary

A guard reduces the annual probability of a robbery by roughly 40% of a 7 pp base. With unguarded neighbors within 500 m, about half of this reduction is offset by displaced robberies.

Displacement policies

From estimates to equilibria

- The estimates pin down the guard premium: $\pi(0) = -\beta_1$ and $\pi(N) = -(\beta_1 + \beta_3)$.
- With an annual cost of EUR 40,000 per guard, Proposition 3 implies that
  - an equilibrium with no guards exists if L_1 is less than EUR 1.37 million
  - an equilibrium with all banks guarded exists if L_N is more than EUR 662,000
- A no-guard equilibrium is likely to exist. In certain highly urban markets, an all-guard equilibrium is also likely to exist.
- Without data on c_i and L_i, the authors cannot identify which equilibrium generates greater social benefits. They simulate four counterfactual scenarios using 2005 data.

Counterfactual scenarios

Scenario	Simulated effect
1. Banning guards	At most five additional robberies in much of the country, more than fifty in Rome, Naples, Milan and Palermo
2. Requiring guards	Robberies fall by over 75% in almost the entire country
3. Gradual removal	The 500 least effective guards deter fewer than ten robberies a year. Each removal raises expected losses by EUR 250 to 1,000
4. Gradual addition	Each additional guard has a small effect
Bank-level approach	Moving guards to markets with few unguarded branches reduces robberies

The simulations ignore costs of robberies beyond the amount stolen, such as the perceived safety of employees and customers.

Conclusion

Conclusion

- Security guards deter bank robberies: a guard lowers the probability of a robbery by 35% to 40%.
- Over half of this reduction is displaced to nearby unguarded banks. Nearby guarded banks receive no displaced robberies.
- Displacement can generate coordination games among banks, so a negative externality does not determine whether policy should encourage or discourage security investments.
- Quantity regulations suit corner equilibria better than price regulations.
- Guards should be discouraged in sparsely populated areas, and large cities may consider encouraging them.

Assessment

Strengths

- Complete data on all potential targets, attempted robberies and security investments.
- The empirical specification follows directly from the model.
- The policy implication of an externality is shown to be ambiguous.

Limitations

- Markets are squares of fixed size, not the choice sets of robbers.
- Guards are observed only at the end of each year.
- Costs c_i and losses L_i are unobserved, so welfare cannot be computed.
- Displacement to other crimes or to more distant targets is not measured.
- The model is static.

Questions for discussion

1. Brazil requires at least two guards in every bank branch. Which equilibrium does this rule select, and under which conditions would the model support it?
2. Branch robberies in Brazil fell from 1,903 in 2000 to 37 in 2021, while digital fraud grew. Can this be interpreted as displacement across types of crime?
3. Hiring decisions are modeled as simultaneous. What would a sequential version, with a leader bank, predict?
4. A bank with several branches in the same market internalizes part of the displacement. How would a merger affect the equilibrium?

Sources: Febraban (2022, 2026), Quod (2026).

Part II. Of Housewives and Feminists

Motivation

- In most industrialised societies, women perform more unpaid work in the home and less paid work in the market than men.
- Economists have long viewed this pattern as specialisation according to comparative advantage (Becker, 1973, Gronau, 1973), assuming that women are relatively less productive in the labour market.
- More recent work suggests that gender gaps in the labour market arise only **after** couples begin to divide labour along gender lines (Goldin, 2014, Blau and Kahn, 2017, Kleven et al., 2018).
- If productivity differences are a result rather than the initial reason for specialisation, what other factors lead couples to divide labour along gender lines?
- This paper focuses on **gender norms** and their potential to reinforce the gendered division of labour.

Motivation: the Brazilian context

- Women spend 21.3 hours a week on household chores and care of people, against 11.7 hours for men (2022).
- Labour force participation is 53.3% for women and 73.2% for men (2022).
- Among those aged 25 to 54, the employment level of women with children up to 6 years old is 9.6 pp lower than that of women without children (2022).
- Women's average income corresponds to 78.9% of men's (2022).

Source: IBGE, Estatísticas de Gênero, 3rd edition (2024).

Identifying the effect of norms: empirical challenges

1. The impact of norms on the household division of labour is notoriously difficult to study empirically.
2. Prior work documents behaviours consistent with norms regulating women's involvement in market work (Fernández, Fogli and Olivetti, 2004, Alesina, Giuliano and Nunn, 2013, Bertrand, Kamenica and Pan, 2015).
3. Outside the lab, norms are correlated with gender differences in preferences for market work and in beliefs about the returns to maternal time (Beblo and Görges, 2018, Fortin, 2015).
4. Hence it is hard to separate norms from differences in productivity and preferences.

Approach: a lab experiment that varies the salience of gender norms exogenously and rules out gender differences in productivity and preferences by design.

Related literature and contribution

- **Gender norms.** Akerlof and Kranton (2000) and Bertrand, Kamenica and Pan (2015). Cudeville and Recoules (2015) show in a model that norms and social conformity can distort labour supply in couples.
  → *Causal evidence that norms affect specialisation choices, independent of other motives.*
- **Focal points.** Schelling (1960), Crawford, Gneezy and Rottenstreich (2008), Isoni et al. (2013). Holm (2000) shows that gender can be a focal point in coordination games. Theory on norms as focal points in the household: Hadfield (1999).
  → *An empirical test of whether gender norms provide a focal point in the household specialisation problem, a question so far studied only in theory.*
- **Experiments with real couples.** Görges (2015) and Cochard et al. (2018) study the division of labour in the lab.
  → *Women claim a smaller share of a joint pie when gender norms are salient.*

This paper

Question: do gender norms affect whether couples reach a male- or a female-favouring division of labour in a non-cooperative setting?

- **Design:** 192 subjects, real heterosexual couples, play a battle of the sexes once with their partner and once with a stranger.
- **Treatment:** the actions are labelled A vs. B (Neutral) or Career vs. Family (Norm).

Main results

- Women choose Career at a significantly lower rate in Norm, with partners and with strangers.
- Men's response is weak and differs between partner and stranger pairings.
- Not explained by beliefs, but consistent with some men signalling progressive attitudes to their partners.

Experimental design

The game (Table 1)

		Player 2: A / Career	Player 2: B / Family
Player 1	A / Career	$0,\ 0$	c, f
	B / Family	$f,\ c$	$0,\ 0$

$c > f > 0$, with $c = 200$ and $f = 100$ ECU (1 ECU = 1 euro cent).

- Total payoffs are highest when one partner chooses Career and the other Family, reflecting gains from specialisation.
- Payoffs favour the partner who chooses Career, who may control the labour income or be better off in case of divorce.
- Payoffs are symmetric across the two equilibria and constant across treatments, so only the labels change.

Equilibria and hypotheses

- Two pure-strategy Nash equilibria: (Career, Family) and (Family, Career).
- In the mixed-strategy equilibrium, the probability of coordination is just below 45%.
- Under standard game theory, the labels of the actions should not affect behaviour.

Focal point hypothesis (Schelling, 1960)

When the actions are labelled Career and Family, the culturally dominant **Traditional** equilibrium (man Career, woman Family) should be reached more often. This improves coordination at the cost of higher inequality to the disadvantage of women.

- Comparing partners with strangers shows whether couple-specific norms can override societal norms.

Why a one-shot non-cooperative game?

- **No communication.** Communication would make the game cooperative for some couples but not for others, and would make the comparison between partners and strangers meaningless.
- **Non-cooperative behaviour is common in couples.** Many experiments with couples document it (Ashraf, 2009, Iversen et al., 2011), and it matters at least for the threat point (Lundberg and Pollak, 1994).
- **One shot.** The first specialisation decision is likely to predict later ones, since wage profiles diverge once one partner focuses on the career.

In-class exercise

Exercise: the specialisation game

Two partners simultaneously choose Career (C) or Family (F), without communicating. Payoffs are (Partner 1, Partner 2), with $c > f > 0$:

		Partner 2	
		C	F
Partner 1	C	$0,\ 0$	c, f
	F	$f,\ c$	$0,\ 0$

In the experiment, $c = 200$ and $f = 100$.

Exercise: questions

(a) Find all pure-strategy Nash equilibria. Does any player have a dominant strategy?

(b) Find the mixed-strategy equilibrium and each player's expected payoff. Compare them with the pure-strategy equilibria.

(c) Show that the probability of coordination in the mixed equilibrium is $2cf/(c+f)^2$. Evaluate it for $c = 200$ and $f = 100$. How does it change as c increases?

(d) Partner 1 believes that Partner 2 chooses Career with probability q. For which values of q is Family a best response?

Exercise: the data

Shares of subjects choosing Career in the experiment:

	Neutral	Norm
Women	0.56	0.25
Men	0.54	0.62

(e) Assuming that choices are independent, compute the probability of the Traditional equilibrium, the Progressive equilibrium and no coordination in each treatment.

(f) Compute women's expected payoff as a share of men's in each treatment, and compare it with the 67% reported in the paper for the Norm treatment.

Shares implied by Table 4, column (1), with partner and stranger pairings pooled.

Procedure and sample

Procedure

- Conducted at the economics lab of the University of Hamburg in 2016 and 2017 (z-Tree), at the end of 9 sessions of a larger experiment on the division of labour.
- 80 subjects in Norm (4 sessions) and 112 in Neutral (5 sessions). The treatment varies between subjects, the pairing within subjects.
- Each subject plays once with the real partner and once with a randomly matched stranger of the same sex as the partner, in random order. Outcomes are revealed only at the end.
- Partners sit in closed booths and hand in their phones. Choices are revealed to the partner after the experiment.
- After each choice, subjects predict the other player's choice and the number of men and women choosing Career (incentivised).

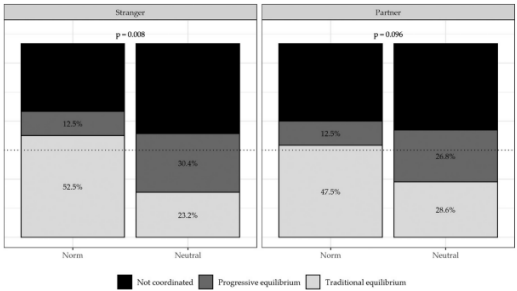
Sample (Table 2)

	Neutral ($N = 112$)	Norm ($N = 80$)	p-value
Age (years)	24.75	27.09	0.002
Union duration (years)	3.00	3.00	0.995
Cohabiting	0.27	0.28	1.000
Married	0.07	0.00	0.038
Satisfaction with union (0 to 10)	8.83	8.53	0.238
Undergraduate student	0.52	0.46	0.542
Ever studied economics	0.33	0.45	0.125
Ever participated in an experiment	0.46	0.71	0.001
Risk aversion (0 to 10)	5.24	5.18	0.850

Selected rows. Joint test of balance: $F(18, 173) = 2.274$, $p = 0.003$. Age, marriage and lab experience are used as controls.

Main results

Coordination outcomes (Figure 2)



- In Neutral, Traditional and Progressive equilibria occur at similar rates.
- In Norm, the Traditional equilibrium is reached more than twice as often.
- Miscoordination falls in Norm, but less than expected from a focal point (dotted line: 45% benchmark).
- χ^2 test: significant for strangers, marginal for partners.

Source: Görges (2021), Figure 2

Coordination outcomes (Table 3)

	Progressive	Not coordinated
Norm	-1.805^{***} (0.69)	-1.155^{**} (0.51)
Partner	-0.334 (0.52)	-0.248 (0.47)
Norm $\times$ Partner	0.434 (0.88)	0.484 (0.68)
Average marginal effect of Norm	-0.171^{***} (0.06)	-0.085 (0.08)

Multinomial logit, Traditional as reference outcome, female sample, with controls (column 2). AME of Norm on Traditional: 0.255^{***} (0.075). Robust standard errors in parentheses.

- Norm lowers the probability of the Progressive equilibrium by about 17 pp and raises that of the Traditional one by about 25 pp.
- Miscoordination falls by about 9 pp, which is not significant.
- In Norm, women earn 67% of men's income. In Neutral, both earn about the same.

Individual Career choices (Table 4)

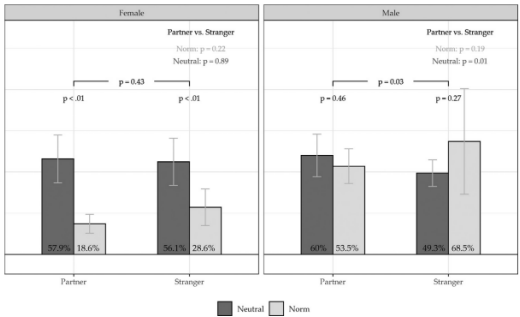
	(1)	(2)	(3)
Norm	0.089	0.217	0.192
Female	0.027	0.071	0.068
Norm × Female	-0.402^{***}	-0.471^{**}	-0.467^{**}
Partner		0.107^{***}	0.107^{***}
Norm × Partner		-0.257^{**}	-0.257^{**}
Female × Partner		-0.089	-0.089
Female × Norm × Partner		0.139	0.139
Intercept	0.536^{***}	0.482^{***}	0.423^{**}
Controls	No	No	Yes
N	384	384	384

- 53.6% of men choose Career in Neutral. The share rises by about 9 pp in Norm, which is not significant.
- For women, the difference in differences is about -40 pp: their probability of choosing Career falls by more than 30 pp in Norm.

Linear probability model with session random intercepts and robust standard errors (omitted here).

Dependent variable: choosing Career.

Career choices by gender and pairing (Figure 3)



- **Women** in Neutral choose Career more than twice as often as in Norm, with partners and with strangers ($p < 0.01$ in both).
- **Men** show no effect of norms with their partner ($p = 0.46$), and an increase of almost 20 pp with strangers that is not significant ($p = 0.27$).
- The difference in the men's response between pairings is significant ($p = 0.03$).

Source: Görges (2021), Figure 3

Mechanisms

Differences in beliefs? (Section 5.1)

- Men's predictions of women's Career choices do not differ between Neutral and Norm, so men overestimate women's Career choices in Norm.
- In Norm, men expect their partner to choose Career less often than a stranger (33% vs. 66%, $p = 0.01$). Beliefs therefore cannot explain why men choose Career less often with their partner.
- Women predict higher rates of Career among men in Norm than in Neutral, consistent with their own choices.

Conclusion: beliefs explain women's choices, but not the heterogeneity in men's response across pairings.

Men “acting feminist”? (Section 5.2)

- Following Bursztyn, Fujiwara and Pallais (2017), men in non-committed relationships may signal progressive gender attitudes, which women value in a long-term partner.
- Relationships are defined as committed when partners cohabit.

Non-committed men choosing Career	Neutral	Norm	p-value
Paired with the partner	54%	33%	0.07
Paired with a stranger	48%	78%	0.01

- No such differences for committed men or for women.
- Caveats: the committed subgroups are small (30 and 22 subjects), and the experiment was not designed to test this hypothesis.

Conclusion

Conclusion

- Gender norms affect coordination outcomes: the Traditional equilibrium is reached significantly more often in Norm, which increases payoff inequality to the benefit of men.
- Women are not compensated by time with children or by redistribution within the couple, which are absent by design. The trade-off between efficiency and equality in the family is a real concern.
- Overall coordination improves only by a small and insignificant margin, because men respond much less than women.
- The heterogeneous responses suggest that norms operate through more complex mechanisms, especially in young couples who may perceive a multiplicity of norms.

Assessment

Strengths

- Only the labels of the actions vary between treatments.
- Real couples, compared with strangers.
- Incentivised beliefs help to discriminate between mechanisms.
- A direct application of the battle of the sexes.

Limitations

- Small and highly selected sample of young couples.
- Low stakes and a stylised, one-shot decision.
- Treatment assigned across only 9 sessions, and the groups are not balanced on observables.
- The analysis of mechanisms is exploratory.

Discussion

The two papers compared

	Maheshri and Mastrobuoni (2021)	Görges (2021)
Players	Banks in a market	Partners in a couple
Game	Coordination game in guard hiring	Battle of the sexes
Multiple equilibria	Few guards or all guards	Traditional or Progressive
Equilibrium selection	Regulation (requirements or bans)	Social norms (focal points)
Evidence	Administrative data, fixed effects	Lab experiment, randomised labels

In both papers, the Nash equilibrium does not pin down the outcome. Which equilibrium is reached is an empirical question.

Questions for discussion

1. Would you expect the salience of gender norms to have a stronger or weaker effect in Brazil? Why?
2. Which policies could change the focal point in the household specialisation problem? Consider parental leave rules.
3. How would the predictions change if partners could communicate before choosing?
4. The division of labour is decided repeatedly over the life cycle. How could a repeated game change the analysis?

Presenting a paper: suggested structure

1. **Motivation:** the question and why it matters.
2. **Related literature and contribution.**
3. **This paper:** design and main results in one slide.
4. **Model:** players, strategies, payoffs and equilibrium.
5. **Data or experimental design.**
6. **Empirical strategy:** threats to identification and how the authors address them.
7. **Results:** the main tables and figures.
8. **Conclusion and assessment:** strengths, limitations and open questions.